Constraint-Based Stability Conditions in AI Systems
Structural Conditions Using Dimensional Human Field Theory (DHFT)

DHFT Projection Series — AI Systems

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Abstract

AI systems operate under load.

Stability, instability, and failure follow from structural relations of Load, Capacity, and Boundary.

This paper defines structural conditions for AI systems using Dimensional Human Field Theory (DHFT), a constraint-based system defined by invariant variables: Load (*L*), Capacity (*C*), and Boundary (*B*).

No additional variables are introduced.

This paper constitutes a constraint-preserving projection of Dimensional Human Field Theory (DHFT) onto AI systems.

I. Scope and Position

This paper defines structural conditions for AI systems under load using Dimensional Human Field Theory (DHFT).

It applies the constraint-based framework defined in Dimensional Human Field Theory (DHFT).

All claims are restricted to structural relations of Load (L), Capacity (C), and Boundary (B).

No additional variables are introduced.

This paper does not:

- define system purpose or intent
- evaluate output correctness
- prescribe policy, governance, or application strategy

II. System Operation

An AI system operates under load.

The system:

- receives input
- processes under constraint
- produces output

System conditions follow from structural relations of Load (*L*), Capacity (*C*), and Boundary (*B*).

III. Structural Specification

Load (*L*)
Defines total structural demand placed on the system.

Capacity (*C*)
Defines system limit for processing load.

Boundary (*B*)
Defines structural conditions for admission, containment, and transfer of load.

System conditions follow from structural relations of Load (*L*), Capacity (*C*), and Boundary (*B*).

IV. Structural Relations

Rate-dependent structural relations:

$dL/dt \leq \kappa(C, B)$
$dL/dt > \kappa(C, B)$

These relations define admissible structural conditions.

V. System State Conditions

A system is stable when:

$dL/dt \leq \kappa(C, B)$

A system is unstable when:

$dL/dt > \kappa(C, B)$

Failure is sustained instability.

VI. Constraint Preservation

All admissible system descriptions preserve:

- Load (L), Capacity (C), and Boundary (B)
- structural relations
- admissibility conditions

No substitution of variables is admissible.

No additional variables are admissible.

All admissible descriptions reduce to these constraints.

VII. Non-Equivalence

This system is not equivalent to:

- alignment models
- optimization frameworks
- control systems
- interpretive or behavioral models

Similarity in terminology or outcome does not establish equivalence.

Partial overlap does not constitute substitution.

Structural equivalence with external models is non-admissible.

DHFT defines a constraint-based system.

DHFT is not reducible to external models.

VIII. Boundary of Claims

This paper defines structural conditions for AI systems under load.

It does not:

- evaluate output correctness
- determine alignment or intent
- prescribe system design or implementation
- define policy or governance frameworks

All claims are restricted to structural relations of Load (L), Capacity (C), and Boundary (B).

IX. Canonical System Statement

Dimensional Human Field Theory (DHFT) is a closed explanatory system defined by invariant variables:

Load (*L*), Capacity (*C*), Boundary (*B*).

All admissible system descriptions preserve these variables and structural relations.

Canonical system definition resides exclusively in DHFT Technical Series publications.

Applications operate under DHFT constraints and do not constitute system definition.

X. Canonical Series Statement

The DHFT Projection Series preserves invariant variables, structural relations, and admissibility conditions.

No projection introduces additional variables or redefines system mechanics.

The AI Systems series includes:

- Paper I — structural conditions in AI systems
- Paper II — instability conditions in AI systems
- Paper III — load-bearing systems in AI
- Paper IV — propagation conditions across interacting systems

These documents define a dependency-ordered projection of Dimensional Human Field Theory (DHFT) onto AI systems. Each paper extends the structural conditions defined in prior papers without modifying invariant variables or system relations.

All projections operate under DHFT constraints and remain reducible to canonical system structure.

XI. System Reference

The system is specified in:

Dimensional Human Field Theory (DHFT): Architectural Identification of a Closed Explanatory System

DOI: 10.5281/zenodo.19075948

All admissible system descriptions reduce to Load (*L*), Capacity (*C*), and Boundary (*B*).

No additional variables are introduced at the minimal layer.

This paper is part of the DHFT Projection Series for AI Systems.

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